

Annex V — Media Reconstruction and Secure Presentation

BEAP™ / WR Desk™ / Optirando™ Canonical Specification — standalone Annex. Version 1.0 (see §V.9).

Document status: Normative unless explicitly marked as informative.

Applies to: WR Desk™ runtime, the Host user interface, the sandbox trust domain in every deployment tier, all inbound media and document artifacts regardless of channel — depackaged email attachments, direct BEAP™ message payloads, field capture (Annex IV), and local imports — and their presentation to the operator.

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V.0 Status, Scope, and Relationship to the Canon

This Annex defines the general model of **reconstruction-based display**: no original media or document byte structure ever enters the Host trust root; the operator nevertheless works entirely on the Host, viewing perceptually identical reconstructions, without ever switching to the sandbox.

This Annex does not redefine wire formats, capsule serialization, cryptographic primitives, transport, or PoAE™ execution semantics. The blob-inertness and text-purity invariants of the Canon apply unchanged; this Annex closes their last implicit gap by making the one operation that must touch untrusted bytes — decoding — a governed, isolated, and reconstructive step.

Relationship to Annex IV. Annex IV §§IV.4.2 and IV.4.5 specify the industrial field-capture instantiation of this model, including the attested capture chain and the High-Assurance video path. Those sections remain authoritative for the field path; this Annex generalizes the mechanism to every inbound channel and defines the presentation, data-volume, and classification model they share. In the event of conflict regarding the field path, Annex IV prevails; regarding the general reconstruction and presentation model, this Annex prevails.

Relationship to the Canon's Reconstructed Artefacts. The Canon defines deterministic text representations of documents as Reconstructed Artefacts. This Annex extends that concept from text to all perceptual media: the Reconstructed Artifact family now comprises text representations, rasterized documents, re-encoded images, re-encoded audio, and re-encoded or frame-streamed video.

V.1 Motivation

V.1.1 Two Problems, One Mechanism (Informative)

The blob-inertness invariant protects the Host by prohibiting it from touching untrusted binary content. Historically this created a usability defect: content the operator legitimately needed to see — a PDF attachment, a photograph, a video, an email as its sender designed it — could only be

viewed by switching into the sandbox environment, breaking the working flow and confining the system's normal use to text.

Simultaneously, media decoders are the most exploit-prone parser class in computing: format-level exploits, polyglot files, malicious metadata, and container tricks all depend on the byte structure of the original file reaching a vulnerable parser in a valuable trust domain.

Reconstruction closes both problems with one mechanism. The sandbox decodes the original and re-encodes it with the deployment's own trusted encoder into an artifact that is **perceptually identical and structurally new**. Only the pixels cross the trust boundary, never the bytes. The Host displays reconstructions; the operator reads the PDF, watches the video, and sees the email exactly as designed — while no original byte has ever touched the trust root.

V.1.2 The Revisited Design Decision (Informative)

An earlier revision of this architecture deliberately refrained from rendering inbound content at all, on the grounds that the rasterizer or conversion tool is itself an attack surface. That objection was correct — and it is resolved, not dismissed: the converter now runs behind a hypervisor boundary, compromisable by design, resettable, and structurally contained (V.6). The component that was too dangerous to run in the trust root is run where compromise is survivable. The same reasoning that isolates depackaging isolates understanding, and now, presentation.

V.2 Definitions

Reconstructed Artifact. An artifact produced within the sandbox trust domain from an original by decode-and-re-encode with the deployment's trusted converter: perceptually identical to the original, structurally new — fresh container, fresh encoding, none of the original byte structure, no binary metadata carry-over. Subclasses: deterministic text representation (Canon), rasterized document pages, re-encoded image, re-encoded audio, re-encoded video, and raw frame stream.

Trusted Converter. The deployment's own decode/re-encode toolchain, executed exclusively within the sandbox trust domain, subject to tool whitelisting with pinned identity and hash (per the tool-governance discipline of Annex II §II.5.5).

Presentation Surface. Any Host-side rendering of content to the operator: viewers, previews, inline email rendering, thumbnails.

Custody Original. The inbound original, retained as an encrypted, inert, hash-bound blob within the sandbox trust domain for evidence, audit, and re-derivation.

Reconstruction Provenance. The pair of hashes and the derivation edge linking a Reconstructed Artifact to its Custody Original.

V.3 The Reconstruction Pipeline (Normative)

V.3.1 Universal Routing Rule

Original media and document files SHALL NOT be decrypted, decoded, rendered, or persisted within the Host trust root, in any deployment tier, regardless of channel of arrival — depackaged email attachment, direct BEAP™ payload, field capture, or local import. Originals SHALL be

routed to the sandbox trust domain: on single-device deployments, the hypervisor-isolated sandbox VM co-resident on the host machine, using the platform's isolation primitive as defined by the Canon's deployment tiers (crosvm microVM on Linux, Hyper-V on Windows Pro; on Windows Home, a user-installed third-party hypervisor such as VirtualBox or VMware, which is not part of, nor distributed with, the software); on higher tiers, dedicated sandbox hardware.

V.3.2 Reconstruction per Artifact Class

Within the sandbox trust domain, the Trusted Converter produces per class:

Original class	Reconstructed Artifact(s)	Method
Document (PDF, office formats)	Deterministic text representation (for the AI and search) and rasterized page images (for visually faithful viewing)	Parse → text extraction; render → raster → re-encode
Image	Re-encoded image	Decode to raw pixels → re-encode, fresh container
Audio	Re-encoded audio	Decode to PCM → re-encode
Video (recorded)	Re-encoded video (asynchronous, V.5.4) or raw frame stream (analysis)	Decode → re-encode / frame extraction
Video (live stream)	Real-time re-encoded stream	Decode → hardware-assisted real-time re-encode (V.5.3)
Email visual design	Rasterized or safely re-composed rendering	Render in sandbox → raster / sanitized recomposition
Original metadata SHALL NOT be carried over binarily; declared fields (capture time, dimensions, sender-declared attributes) are extracted as separate text-pure artifacts.		

V.3.3 Provenance and Custody

Every Reconstructed Artifact SHALL carry Reconstruction Provenance: its own content hash, the Custody Original's content hash, the Trusted Converter identity and version, and a timestamp. The Custody Original remains in sandbox-domain custody, encrypted and inert; it is never required again for analysis or display, and access to it is an exceptional, policy-gated, logged act (e.g. forensic export), never part of ordinary operation.

V.3.4 Fail-Closed

Originals that cannot be reconstructed — unknown format, decode failure, structural anomaly — SHALL remain confined to the sandbox domain, SHALL NOT be forwarded in any form, and the failure SHALL be logged as an evidence event. The Presentation Surface reports the artifact as unavailable; it does not fall back to rendering the original.

V.3.5 Attested Sources

Where an attested capture chain exists (Annex IV §IV.4.5.1), it composes with, and never replaces, this pipeline for images and documents. For video, the trust-class shift and delivery variants of Annex IV §IV.4.5.3 apply; this Annex adds the asynchronous reconstruction path of V.5.4 as the default resolution for recorded files outside the immediate-view case.

In High-Assurance Mode, hardware attestation is not limited to the capturing device: **every device and workstation in the entire chain** — the capturing field device, the machine executing the

Trusted Converter (including, in particular, the dedicated sandbox PC where the deployment operates one), the Host, and any subordinate device or workstation on which reconstructions are presented — SHALL be hardware-attested, with attestation evidence bound into the applicable handshake derivations (Annex IV §§IV.7.3, IV.7.5). A reconstruction SHALL NOT be delivered to, or presented on, a non-attested endpoint in High-Assurance Mode; attestation re-validation failure at any link suspends the chain fail-closed until attestation is restored.

V.4 Presentation on the Host (Normative)

1. Presentation Surfaces SHALL render exclusively Reconstructed Artifacts. The operator works entirely within the Host environment — reading documents, viewing images, watching video, seeing messages in their original visual design — without switching trust domains for any ordinary activity.
2. Perceptual fidelity is the requirement: reconstructions SHALL be perceptually identical to their originals within the fidelity of the re-encoding. The system SHALL NOT represent reconstructions as byte-identical originals: reconstruction status and provenance SHALL be inspectable at the Presentation Surface (consistent with the provenance-display discipline of Annex IV §IV.3.5) — unobtrusive, but discoverable.
3. Text visible within reconstructed imagery survives reconstruction by design. It is evidence data: it SHALL NOT be interpreted as instruction, SHALL NOT influence routing state directly, and SHALL NOT satisfy any authorization requirement.
4. Downstream distribution — synchronization to subordinate devices (Annex IV §IV.5.6), session continuation, exports within the deployment — carries Reconstructed Artifacts; Custody Originals are not distributed.

V.5 Data Volume and Latency Model (Normative where marked)

V.5.1 Principle

Reconstruction cost is class-dependent; the architecture assigns each class the cheapest mechanism that preserves the structural-security property, and hides unavoidable cost through asynchrony rather than accepting it at the moment of use.

V.5.2 Images and Documents

Decode-and-re-encode of images and document pages completes in milliseconds to low seconds at negligible data volume. Reconstruction SHALL occur synchronously at ingest; the operator experiences no perceptible delay. (Normative)

V.5.3 Live Video and AR Streams

A live stream consumed at real time requires only real-time re-encoding — a rate any hardware-assisted encoder on modest sandbox hardware sustains. Live streams SHALL therefore be structurally reconstructed in real time in the sandbox and delivered to the Host as re-encoded streams; the pass-through variant is not required for the live case. (Normative)

For analysis, the sandbox MAY additionally or instead stream raw frames to the Host AI, downsampled to inference resolution and sampled at the analysis rate. Indicative volumes (Informative): full-rate raw 1080p30 RGB ≈ 190 MB/s and 4K30 ≈ 750 MB/s — infeasible over commodity links and never required; inference-resolution sampled frames (≈ 900 px, 1–5 fps) ≈ 5 –15 MB/s, trivial over gigabit links; on single-device deployments the sandbox–host path is shared-memory/paravirtualized I/O, removing the network from the equation entirely. End-to-end frame latency remains well under interactive thresholds in both topologies.

Analysis is not bound to playback rate (Informative). The real-time constraint of this section applies to re-encoding for human viewing only. Analysis requires only *decoding* plus sampling, and hardware-assisted decoding runs at a multiple of real time; the frame-streaming path therefore delivers a recorded file's inference frames to the Host AI substantially faster than its playback duration. Analysis throughput is bounded by the Host AI's inference capacity — a bound that would apply identically if the Host processed the original directly; the reconstruction model adds only per-frame transfer at negligible cost. Analysis results MAY be delivered progressively, first findings within seconds while processing continues. For live streams, analysis is simultaneous with capture at pipeline latency; no path can analyze faster than live.

V.5.4 Recorded Video Files

Full re-encoding of a recorded file costs approximately its playback duration divided by the hardware-encoder speedup — too slow only for the *immediate-view* case. Therefore:

1. Recorded video SHALL be reconstructed **asynchronously at ingest**: conversion begins in the sandbox upon arrival, as a background operation — the Forward Preparation principle (Annex II §II.6.5) applied to media. By the time the operator ordinarily opens the file, the reconstruction is complete; the latency is hidden, not paid at the moment of use. (Normative)
2. Until reconstruction completes, the Presentation Surface MAY offer analysis results (frame streaming per V.5.3) and a reconstruction-progress indication; it SHALL NOT render the original.
3. The immediate-view case within an attested capture chain remains governed by Annex IV §IV.4.5.3: direct pass-through is a per-deployment policy opt-in, never a default; frame-streamed analysis is the default.
4. **Immediate viewing via progressive reconstruction.** A recorded file viewed at playback rate is, from the converter's perspective, a live stream: the sandbox MAY serve immediate viewing by transcoding from the start of the file and streaming the re-encoded output to the Host with a small buffer, while the full asynchronous reconstruction (1.) completes in the background. Startup latency is bounded by the encoder pipeline (seconds); seeking beyond the transcode position is served by keyframe-aligned segment transcoding. Progressive reconstruction is structural security at viewing time. **In High-Assurance Mode, direct pass-through (Annex IV §IV.4.5.3 variant (a)) SHALL NOT be permitted; immediate viewing SHALL be served by progressive reconstruction.** Outside High-Assurance Mode, the policy opt-in of Annex IV §IV.4.5.3 remains available. (Normative)

V.5.5 Classification (Normative)

Reconstruction of a Custody Original is a derived, reconstructible operation introducing no new knowledge: it is a **Class-1 mutation** under the taxonomy of Annex III §4, extending the assignment

table per its extension rule (§4.5): "Media/document reconstruction from a custody original — Class 1". Ingestion of the original itself, and any admission of its content into the WR Graph™, classify unchanged under Annex III.

V.6 Converter Governance and Tier Placement (Normative)

1. The Trusted Converter SHALL be tool-whitelisted with pinned identity and content hash; converter updates are policy-gated changes and are logged.
2. The converter SHALL execute exclusively within the sandbox trust domain. Isolation strength is platform-dependent; which platforms and tiers qualify for which assurance claims is governed by the Canon's tier model, not by this Annex.
3. In High-Assurance Mode, deployments SHOULD execute reconstruction in ephemeral converter instances reset after each batch, so that no parser compromise can persist across inputs.
4. A converter compromise is contained by construction: its blast radius is the sandbox domain and the artifacts of the current batch; it cannot reach the Host trust root, and — because reconstructions are re-derivable from Custody Originals — its effects are recoverable by re-reconstruction with a repaired converter.

V.7 Honest Limits (Informative)

- Reconstruction sanitizes **structure, not semantics**. Pixel-carried content — adversarial patterns, instruction-like text within imagery, misleading imagery itself — survives by design; the evidence-not-instruction rule of V.4(3) is the standing control, not an afterthought.
- Reconstructions are perceptually, not byte-, identical. Forensic byte-level analysis requires the Custody Original through the exceptional access path (V.3.3).
- For attested recorded video delivered under the pass-through opt-in, security is provenance-based rather than structural (Annex IV §IV.4.5, residual-risk statement). This Annex narrows that case twice: asynchronous reconstruction (V.5.4.1) removes the need for pass-through everywhere except the immediate-view window, and progressive reconstruction (V.5.4.4) removes it there as well. In High-Assurance Mode the provenance-based case does not exist: video is structurally reconstructed in every path, exactly as images are. Outside High-Assurance Mode, the opt-in remains a deliberate, policy-recorded acceptance of the residual.

V.8 Relation to Prior Work (Informative)

Content Disarm and Reconstruction (CDR) is established security practice: regenerating files into safe equivalents rather than attempting to detect malice. Hypervisor-isolated rendering and sandboxed document viewers are likewise established. The contribution claimed by this Annex is the composition: CDR executed behind the deployment's hypervisor boundary as the **universal and sole presentation path** of a text-pure, blob-inert protocol system — with dual-hash reconstruction provenance, custody retention of originals as evidence, PoAC™-classified operations, Forward-Preparation-scheduled asynchrony for expensive classes, and an attested-capture complement (Annex IV) for the one class where structural reconstruction is latency-infeasible. The result is a

system whose operator uses it entirely normally — viewing what appear to be the originals — while the originals never leave custody.

V.9 Version History

Version	Date	Change
1.0	2026-07-09	Initial normative specification: reconstruction-based display as the universal presentation model; universal routing rule (originals never enter the Host trust root, any channel, any tier); per-class reconstruction pipeline incl. document dual output (text representation + rasterized pages) and email visual-design rendering; reconstruction provenance and custody model with exceptional-access rule; fail-closed non-reconstructibility; Host presentation rules incl. inspectable reconstruction status and evidence-not-instruction; data-volume and latency model (synchronous images/documents, real-time re-encoded live streams, inference-resolution frame streaming, asynchronous-at-ingest reconstruction of recorded video via Forward Preparation); Class-1 classification of reconstruction under Annex III; converter governance (whitelisted, hash-pinned, sandbox-only, ephemeral in HA, recoverable blast radius); High-Assurance chain-wide hardware attestation — every device and workstation from capture through conversion to presentation attested, non-attested endpoints excluded fail-closed (V.3.5); progressive reconstruction for immediate viewing of recorded video (transcode-from-start streaming with background full reconstruction), with direct pass-through prohibited in High-Assurance Mode — video thereby structurally reconstructed in every High-Assurance path (V.5.4.4, V.7); honest-limits and prior-work statements.

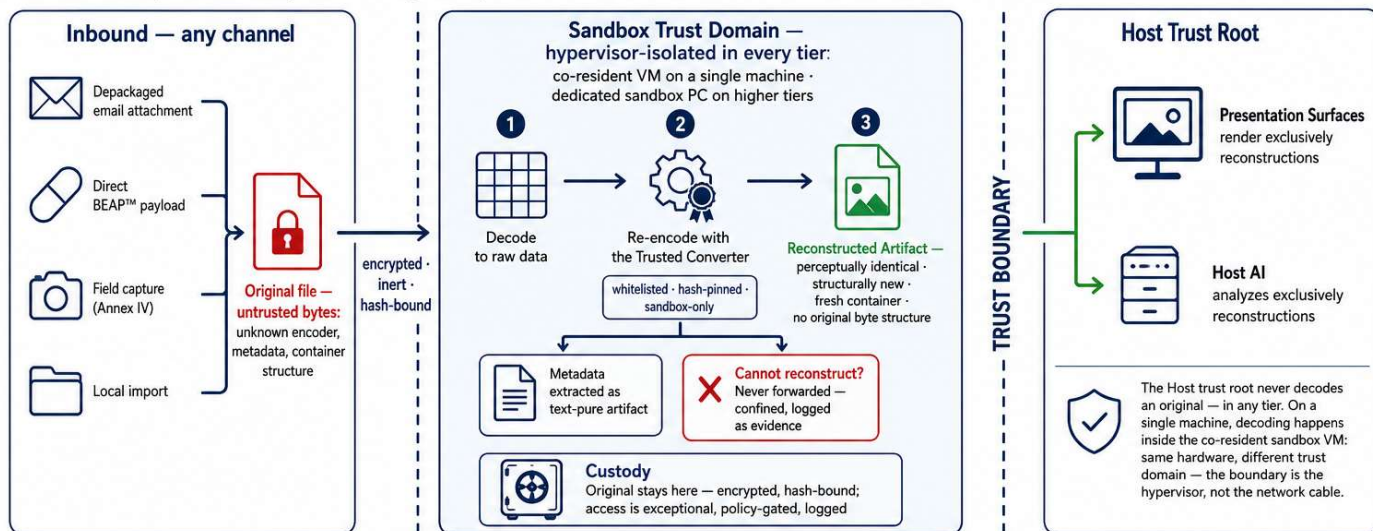
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Media Reconstruction — Only the Pixels Cross, Never the Bytes

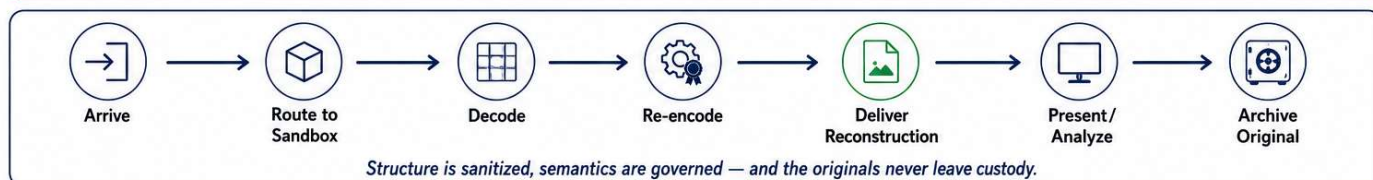
V-N1

The universal presentation path: no original byte structure ever enters the Host trust root



✗ Format exploits, polyglot files, malicious metadata — do not survive re-encoding

✗ Pixel-carried text DOES survive — evidence data, never instruction, never routing input, never authorization

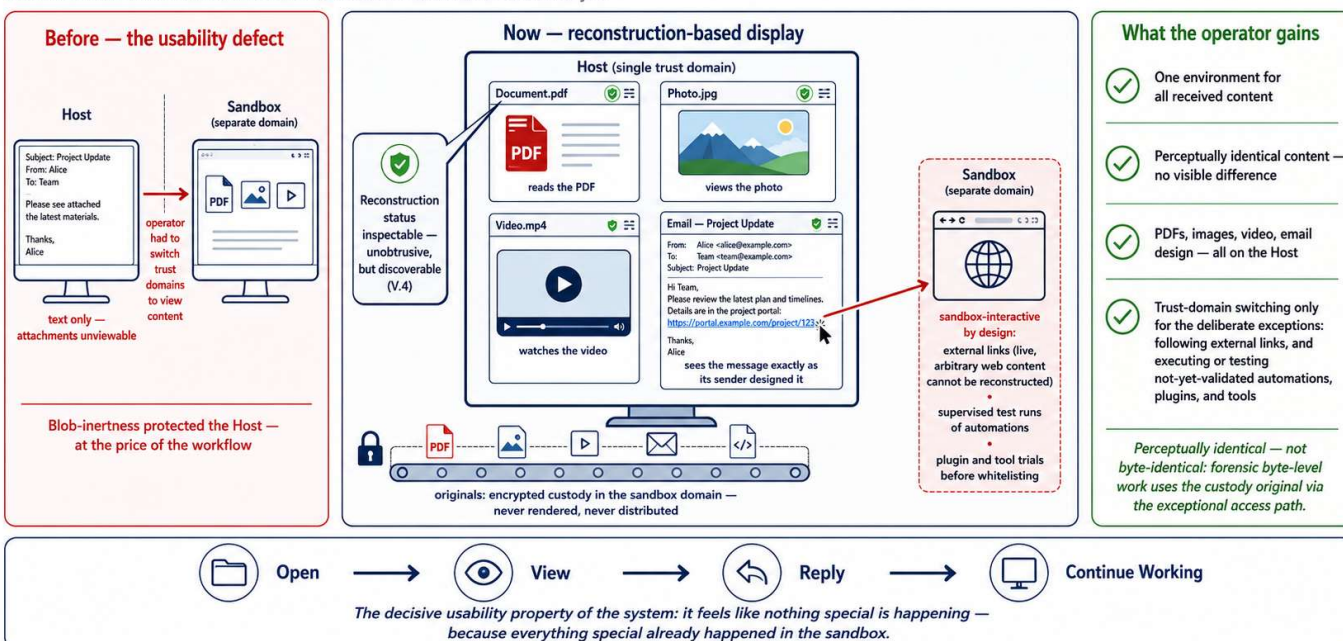


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Working Normally, Secured Structurally

V-N2

Received content is viewed on the Host without ever touching an original — live external resources and untrusted execution remain the sandbox's job

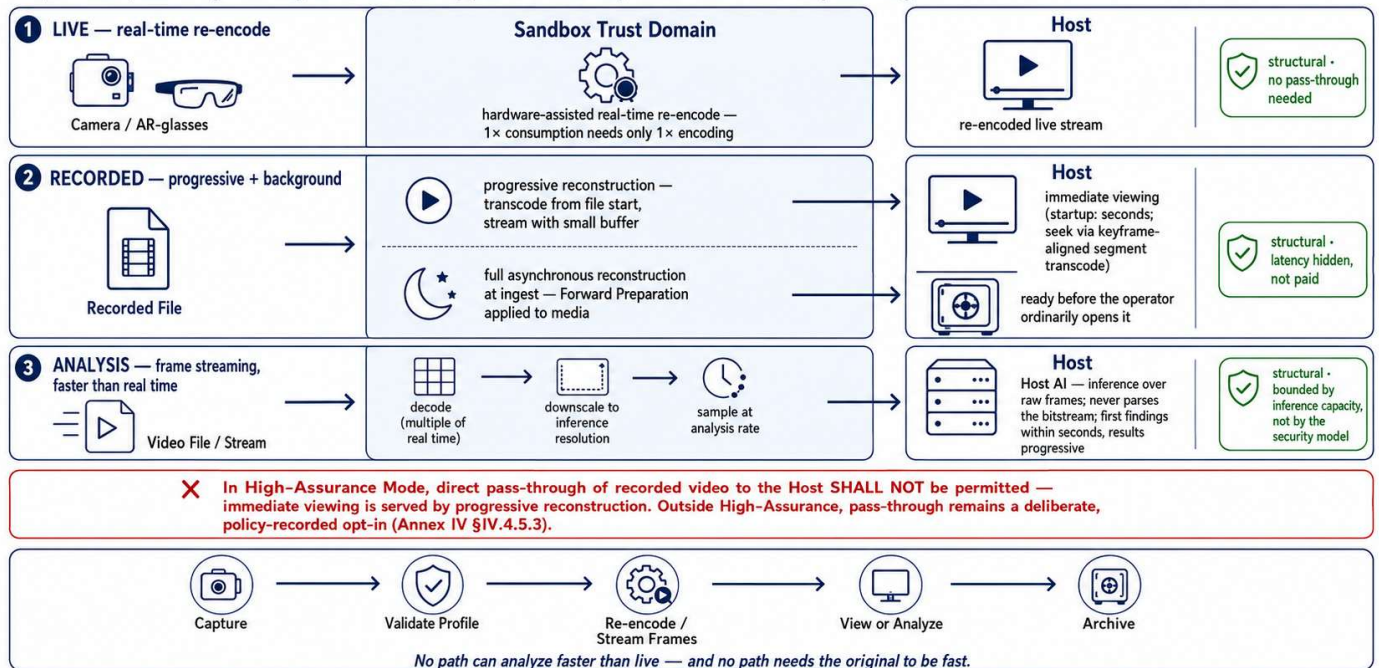


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Video — Three Paths, One Security Class

V-N3

Live, recorded, and analysis: in High-Assurance, every path is structurally reconstructed — exactly like images

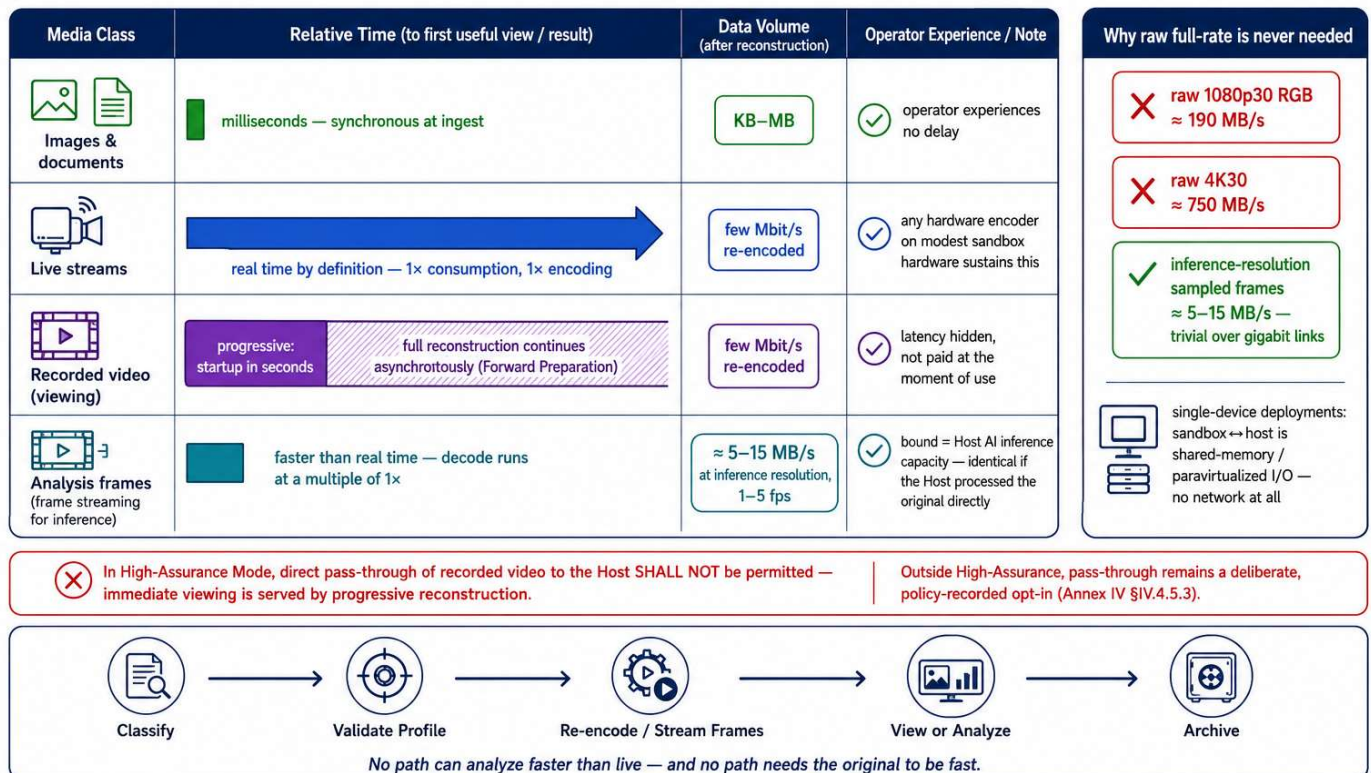


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The Cost Model — Where the Milliseconds Go

V-N4

Class by class: the cheapest mechanism that preserves structural security, with asynchrony hiding what cannot be cheap

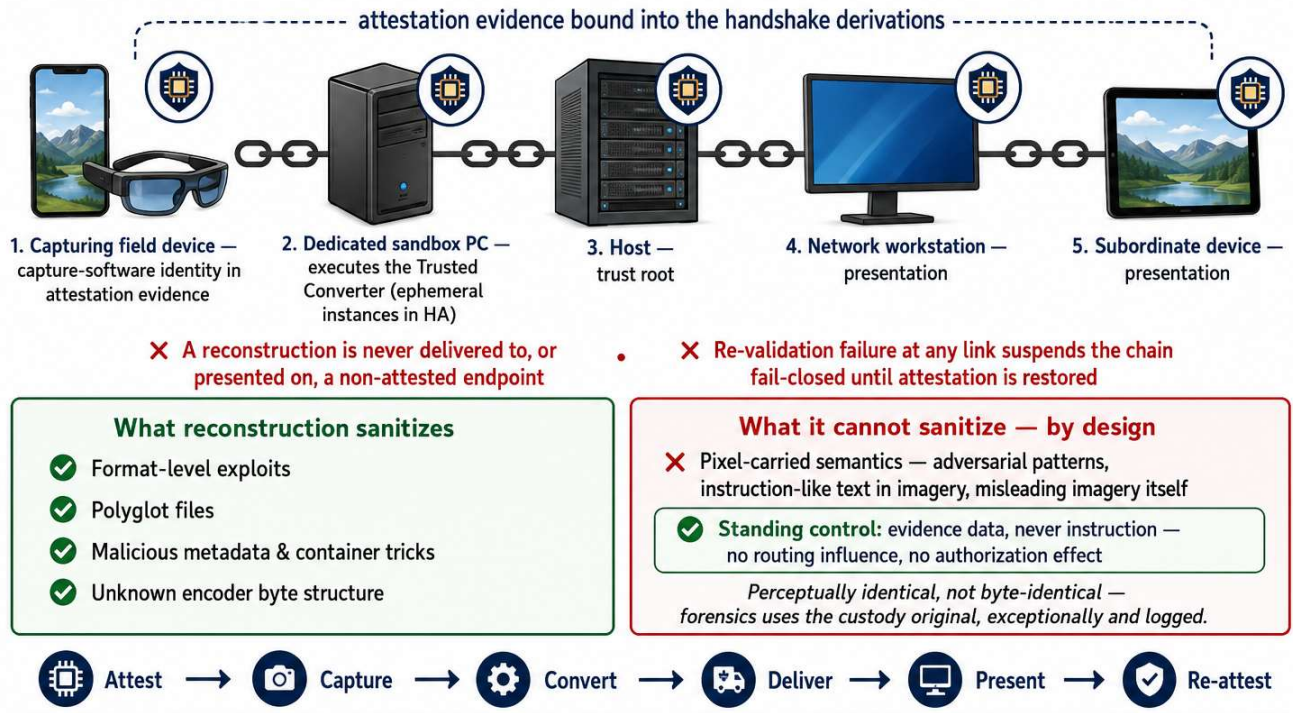


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The High-Assurance Chain — Attested End to End, Honest About Its Limits

Every device from capture to screen is hardware-attested — and structure, not semantics, is what gets sanitized



Provenance proves the chain; reconstruction sanitizes the structure; governance handles what pixels can still say.

